

Software Test Plan (STP) – Bank Management System

Project: Bank Management System

Field	Details
Version	1.0
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1. Introduction

Purpose: This document defines the test plan for the Bank Management System (BMS) v1.0. It outlines objectives, scope, strategy, resources, schedule, responsibilities, and reporting activities for testing.

Scope: Testing covers authentication and user management, account management, balance inquiry, cash deposit, cash withdrawal, fund transfer, beneficiary management, transaction history, account statements, loan management, notifications, administration, reporting, audit logging, security controls, performance, and availability.

References: Bank Management System SRS v1.0, Design Specifications v1.0, Requirements Traceability Matrix (RTM), and applicable organizational banking/security requirements.

Definitions: BMS (Bank Management System), SRS (Software Requirements Specification), STP (Software Test Plan), RTM (Requirements Traceability Matrix), OTP (One-Time Password), RBAC (Role-Based Access Control), KYC (Know Your Customer), UAT (User Acceptance Testing), API (Application Programming Interface).

2. Test Items

- Authentication and OTP module
- Customer profile and KYC module
- Account creation and account-status module
- Balance inquiry and transaction-history module
- Cash deposit module
- Cash withdrawal module
- Funds transfer and beneficiary module
- Loan application and repayment module
- Notification module
- Administration and RBAC module
- Reporting and account-statement module
- Audit logging and monitoring module
- Security controls
- Performance and availability behavior

3. Features to be Tested

Features mapped to the BMS SRS requirement IDs:

SRS Requirement ID	Feature	Test Case Reference
BMS-F-001 / F-002	Authentication and OTP	TC-AUTH-01, TC-AUTH-02
BMS-F-003	Customer profile and KYC	TC-CUST-01
BMS-F-004 / F-005	Account creation/status	TC-ACC-01, TC-ACC-02
BMS-F-006	Balance inquiry	TC-BAL-01
BMS-F-007	Cash deposit	TC-DEP-01
BMS-F-008	Cash withdrawal	TC-WD-01
BMS-F-009 / F-010	Funds transfer/beneficiaries	TC-TRF-01, TC-BEN-01
BMS-F-011 / F-012	History/statements	TC-HIST-01, TC-STMT-01
BMS-F-013 to F-016	Loan management	TC-LOAN-01 to TC-LOAN-04
BMS-F-017	Notifications	TC-NOTIF-01
BMS-F-018	Administration/RBAC	TC-ADMIN-01
BMS-F-019	Operational reports	TC-REP-01
BMS-F-020	Audit trail	TC-AUD-01
BMS-NF-001	Response time $\leq$ 3 seconds target	TC-PERF-01

BMS-NF-002	99.9% availability target	TC-OPS-01
BMS-NF-003 / BMS-SR-001	Encryption/TLS	TC-SEC-01
BMS-SR-002	Secure password storage	TC-SEC-02
BMS-SR-003	RBAC enforcement	TC-SEC-03
BMS-SR-004	Failed-login lockout	TC-SEC-04
BMS-SR-005	Audit/security logging	TC-SEC-05

4. Features Not to be Tested

- Internal implementation of the relational database engine.
- Operating-system internals and physical server firmware.
- Third-party SMS/email provider internal processing after BMS handoff.
- External bank/inter-bank network internal processing beyond the BMS API boundary.
- Production banking data and real customer credentials.
- Infrastructure performance outside the agreed BMS test environment.

5. Test Approach / Strategy

5.1 Levels

- Unit tests (module-level).
- Integration tests (BMS modules, database, authentication/OTP and API boundaries).
- System tests (end-to-end banking workflows).
- Acceptance tests (UAT).

5.2 Types

- Functional testing
- Regression testing
- Negative and boundary testing
- Performance testing
- Security testing
- Usability/accessibility testing
- Reliability/recovery testing

5.3 Entry Criteria

- Stable build delivered.
- Test data and user roles available.
- Test environment and database ready.
- Major blocking defects from the previous build resolved or accepted.
- Test cases and RTM reviewed.

5.4 Exit Criteria

- 100% of planned test cases executed.
- All high-priority requirements have coverage.
- 0 critical defects remain open.
- High-priority acceptance criteria satisfied.
- Known lower-priority defects documented and accepted.
- RTM shows requirement-to-test coverage.

5.5 Security Validation

- Validate password handling and absence of plaintext passwords.
- Verify TLS 1.2+ for client-server and external API communication.
- Verify RBAC for customer, staff, administrator, and auditor functions.

- Verify failed-login logout and audit event.
- Verify financial/security actions create audit records without passwords or OTP values.
- Perform negative and boundary input testing.
- Test session handling and authorization boundaries.

6. Test Environment

Area	Configuration
Hardware	QA/developer workstations; application server/local server; database server; optional statement/receipt printer.
Software	BMS v1.0; supported web browser(s); relational DBMS; authentication/OTP test service; notification sandbox.
Tools	Selenium or equivalent UI automation; Postman; JMeter; Git; Jira or equivalent defect tracker.
Test Data	Dummy customers/staff/admin/auditor accounts; valid/invalid accounts; balances; beneficiaries; deposits; withdrawals; transfers; loan data.
Network	Controlled LAN/WAN or local test network with HTTPS/TLS and simulated external endpoints where required.

7. Test Schedule

Milestone	Planned Date
Test plan and test-case design	01-Oct-2026 to 04-Oct-2026
Environment setup	05-Oct-2026 to 06-Oct-2026
Unit/integration execution	07-Oct-2026 to 11-Oct-2026
System/functional execution	12-Oct-2026 to 18-Oct-2026
Security/performance testing	19-Oct-2026 to 21-Oct-2026
Regression testing	22-Oct-2026 to 24-Oct-2026
UAT	26-Oct-2026 to 28-Oct-2026
Final summary/sign-off	29-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Manual and automated test cases
- Test scripts
- Test data
- Test execution logs/evidence
- Defect reports
- RTM with execution status
- Test Summary Report
- UAT results and sign-off record

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	SRIKANTH V REDDY PES2UG24CS519	Prepare the test plan, coordinate test execution, review results, track requirement coverage, and coordinate final test readiness.
Test Engineer	T. MAHALAKSHMI PES2UG24CS558	Design and execute test cases, prepare test data, record test evidence, and report defects.
Developer	SUDHANWA PES2UG24CS531	Support the test environment, analyze failures, fix defects, and participate in defect triage and regression testing.
Product Owner / Bank Representative	SRUJAN PES2UG24CS526	Clarify business requirements, review acceptance criteria, validate UAT results, and provide final test approval/sign-off.

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds and prioritize testing around available modules.
Test environment downtime	Maintain a backup environment/database snapshot where practical.
Incomplete test data	Prepare and validate controlled test data before execution.
OTP/notification service dependency	Use sandbox or stub services and fallback endpoints.
Frequent business-rule changes	Version requirements and update affected tests/RTM.
Duplicate or inconsistent financial posting	Use isolated test accounts, transaction IDs, DB checks, and reconciliation tests.
Late security defects	Run security validation alongside functional testing.

11. Assumptions & Dependencies

- BMS SRS v1.0 is the testing baseline.
- Dummy test accounts and customer data will be provided.
- QA environment and database will be available during execution.
- OTP and notification sandbox services will be available or stubbed.
- Transaction limits, loan rules, and account-state rules will be provided by stakeholders.
- Defect tracking and version-control systems will be accessible.

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment is unavailable for more than 4 hours and no backup is available.
- A blocking defect prevents more than 30% of planned test cases.
- Database/authentication services are unstable enough to make results unreliable.
- A major build change invalidates a substantial portion of the active test suite.

Resume testing if:

- Blocking defects are resolved or formally waived.
- Environment and dependent services are stable.
- New/repaired build passes smoke testing.
- Required test data and configuration are restored and verified.

13. Test Case Management & Traceability

The RTM shall map BMS SRS requirements to one or more test cases and record execution status.

SRS Requirement ID	Requirement Area	Test Case(s)
BMS-F-001 / F-002	Authentication and OTP	TC-AUTH-01, TC-AUTH-02
BMS-F-007	Cash deposit	TC-DEP-01
BMS-F-008	Cash withdrawal	TC-WD-01
BMS-F-009	Funds transfer	TC-TRF-01
BMS-F-013 to F-016	Loan management	TC-LOAN-01 to TC-LOAN-04
BMS-F-018	Administration/RBAC	TC-ADMIN-01, TC-SEC-03
BMS-F-020	Audit trail	TC-AUD-01, TC-SEC-05
BMS-NF-001	Response time	TC-PERF-01
BMS-NF-002	Availability	TC-OPS-01
BMS-SR-001 to SR-005	Security requirements	TC-SEC-01 to TC-SEC-05

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed/blocked/not executed
- Defect count by severity and priority
- Defect density

- Defect aging and closure time
- Requirement coverage
- Regression pass rate
- Performance response-time measurements
- Security test pass/fail results

Reports:

- Daily/periodic execution status
- Defect status and triage report
- RTM/requirement coverage report
- Performance and security test reports
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner / Bank Representative		